Game Loop Architectures

Fixed Timestep vs Variable Timestep

Game loops control how frequently game logic and rendering are updated. Two common architectural approaches are Fixed Timestep and Variable Timestep.

1. Fixed Timestep

In a Fixed Timestep architecture, the game updates logic at a constant interval (e.g., 60 updates per second). Each update step processes the same amount of simulated time.

Advantages:

- Deterministic behavior (consistent simulation results)
- Stable physics calculations
- Easier debugging and testing

Disadvantages:

- Can waste CPU if frame completes early
- Needs interpolation for smooth rendering

2. Variable Timestep

In a Variable Timestep architecture, game updates depend on the time elapsed since the last frame (delta time). Simulation speed adapts dynamically.

Advantages:

- Efficient CPU usage
- Simpler loop structure
- Smooth animation with delta time

Disadvantages:

- Non-deterministic simulation
- Physics instability at low frame rates

Conclusion

Fixed Timestep is preferred for physics-heavy or deterministic games. Variable Timestep is commonly used in simpler or rendering-focused games. Professional engines often combine both approaches in a hybrid model.